Practical No. 7

Aim: To create and manage users, groups, and permissions using AWS Identity and Access Management (IAM) in order to control secure access to AWS services.

Theory:

AWS Identity and Access Management (IAM) is a service that enables you to securely control access to AWS resources. Using IAM, you can create and manage users, groups, and permissions to allow or deny access to specific AWS services and resources.

IAM follows the principle of least privilege — users are granted only the permissions they need to perform their job. Permissions are defined using policies, which can be attached to users or groups. Managed Policies are reusable pre-built policies, while Inline Policies are assigned to a single user or group for one-off situations.

Key Concepts

- Users — Individual IAM identities with specific credentials and permissions
- Groups — Collections of users that share the same permissions
- Policies — JSON documents that define allowed or denied actions on AWS resources
- Roles — Temporary permissions that can be assumed by AWS services or users

Advantages of IAM

- Centralized control over AWS account access
- Fine-grained permissions at the resource level
- No additional charge — IAM is available at no cost within AWS

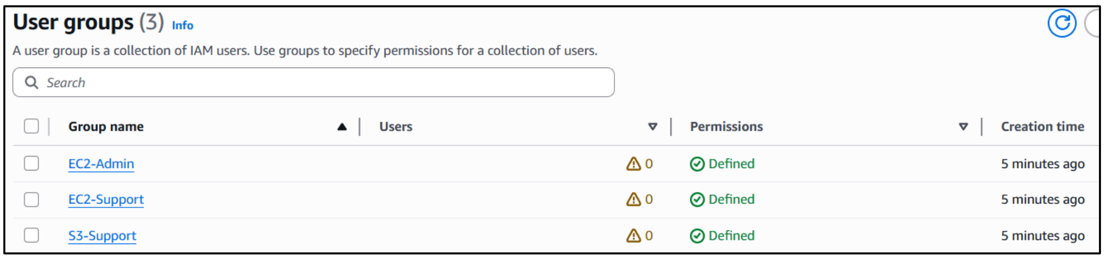
Steps:

Task 1: Explore Users and Groups

1. In the AWS Console, search for and open IAM. In the left navigation pane, choose Users. Three users are available: user-1, user-2, and user-3.

User name	Path	Groups	Last activity	MFA	Password age	Console last sign-in	Access key ID
awsstudent	/	Access denied	-	Access denied	Access denied	-	Access denied
user-1	/spl66/	0	-	-	3 minutes	-	Active - AKIA6GBMFO...
user-2	/spl66/	0	-	-	3 minutes	-	Active - AKIA6GBMFO...
user-3	/spl66/	0	-	-	3 minutes	-	Active - AKIA6GBMFO...

2. Click user-1 and observe that it has no permissions and is not a member of any group. It has a Console password assigned under Security credentials.
3. In the left pane, choose User groups. Three groups exist: EC2-Admin, EC2-Support, and S3-Support.



User groups (3) Info

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

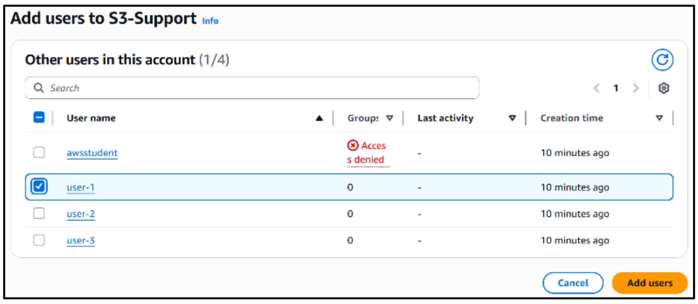
Search

Group name	Users	Permissions	Creation time
EC2-Admin	0	Defined	5 minutes ago
EC2-Support	0	Defined	5 minutes ago
S3-Support	0	Defined	5 minutes ago

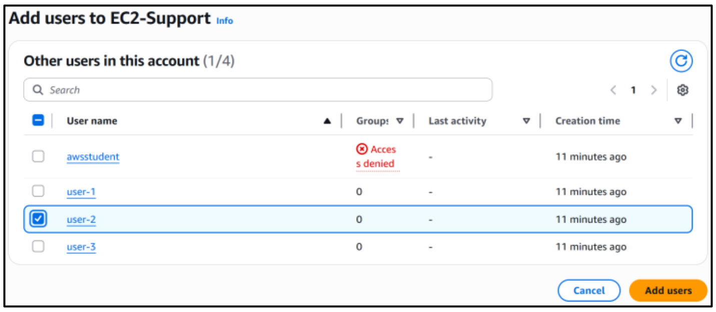
4. Review the Permissions tab of each group. EC2-Support has AmazonEC2ReadOnlyAccess, S3-Support has AmazonS3ReadOnlyAccess, and EC2-Admin has an Inline Policy allowing Describe, Start, and Stop actions on EC2 instances.

Task 2: Add Users to Groups

1. Open the S3-Support group, go to the Users tab, and click Add users. Select user-1 and confirm.



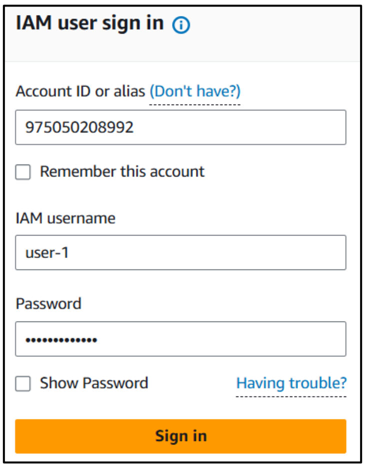
2. Open the EC2-Support group and add user-2 using the same steps.

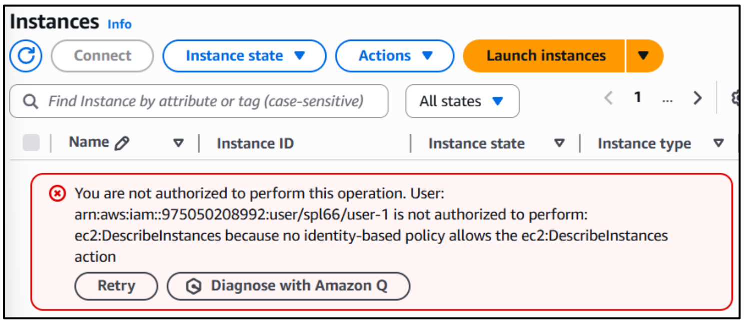


3. Open the EC2-Admin group and add user-3. Verify that each group now shows 1 user in the Users column.

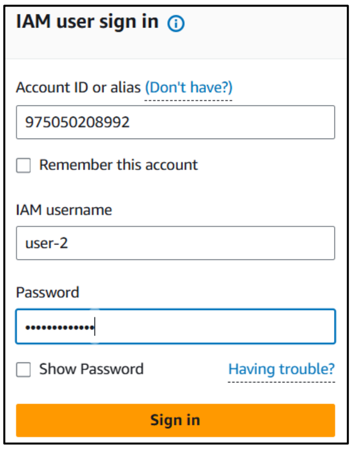
Task 3: Sign In and Test Users

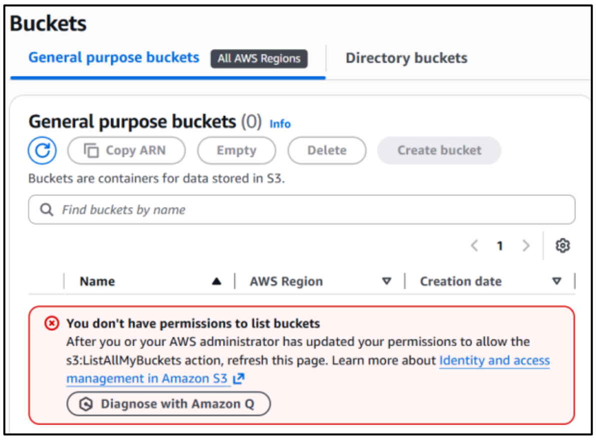
1. From the IAM Dashboard, copy the Sign-in URL for IAM users and open it in a private browser window.
2. Sign in as user-1 (Lab-Password1). Verify that S3 buckets are visible but EC2 access is denied — confirming S3 read-only permissions.

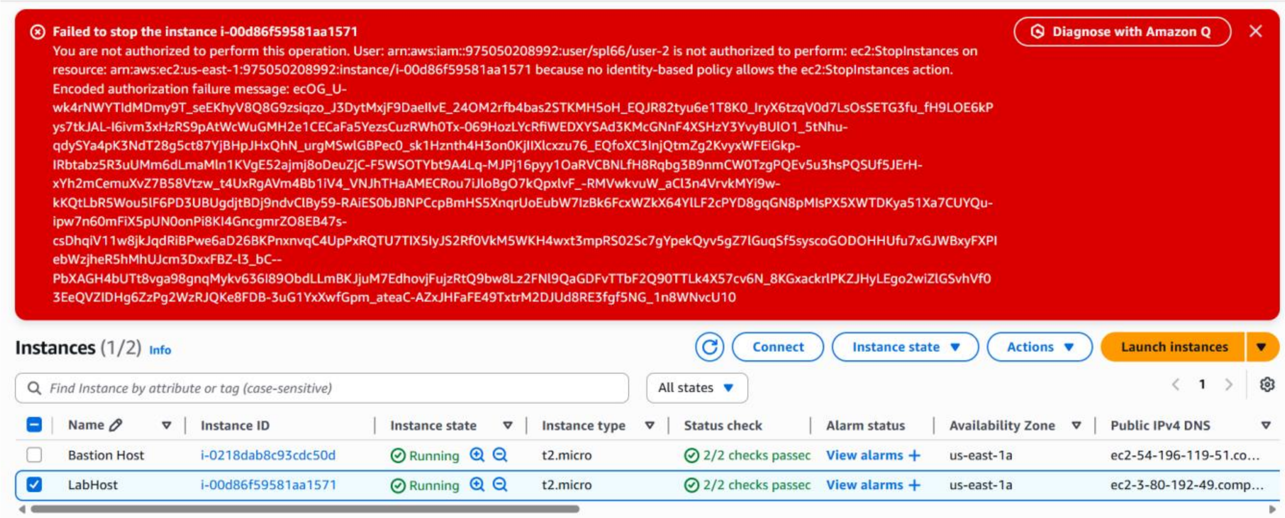




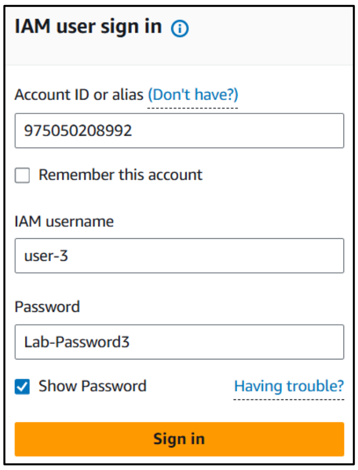
3. Sign out and sign in as user-2 (Lab-Password2). Verify that EC2 instances are visible but stopping an instance is denied — confirming EC2 read-only permissions. Also verify that S3 access is denied.

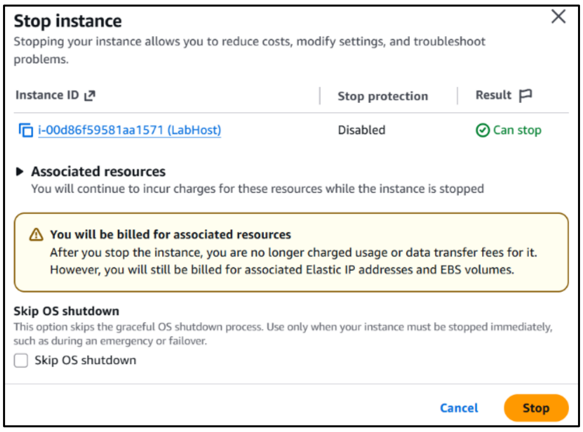






4. Sign out and sign in as user-3 (Lab-Password3). Verify that stopping the EC2 instance named LabHost is successful — confirming EC2 Admin permissions.





Conclusion: Thus, we have successfully created and managed users, groups, and permissions using AWS Identity and Access Management (IAM) to control secure access to AWS services.